

Gracim ban Flamin

912

Page 1

26 June 2020

1. S cannot be empty because there exists at least one solution to $x_{k+2} - (\sin k)x_k = k$, say we know $x_0 = x_1 = 0$.

$$\text{Now } x_2 = \sin 2 \cdot 0 + 2 = 2$$

$$x_3 = \sin 3 \cdot 2 + 3 = 3 + 2 \sin 3$$

$$x_4 = \sin 4 (3 + 2 \sin 3) + 4 = 2 \sin 3 \sin 4 + 3 \sin 4 + 4$$

$\vdots$

$$x_k = 2 \sin 3 \sin 4 \dots \sin k + 3 \sin 4 \dots \sin k + k$$

We also know that for any $x_0 = y_1, x_1 = y_2$ there exists a unique solution because the IVP theorem holds. So we have proven T is injective because any solution is generated by one IVP. To prove surjectivity we need to show that any IVP can generate a solution from S , so there is a one to one correspondence between the IVP and the solution. Thus we have bijectivity.

S is indeed a linear space of finite, in this case dimension 2. We know because any equation of the form $x_{k+n} + a_{1,k} x_{k+n-1} + \dots + a_{n,k} x_k = f_k, k \geq 0$ has solutions which form a linear space where addition and multiplication are defined in a natural way. (from the lecture)

26 June 2020

Graciu Ioan Flavin

group 912

Page 2

2. $x'' + 2tx' = 0$

~~We can integrate the~~

Denote $y = x'$, the system becomes $y' + 2ty = 0$

Remember from lecture 3 the solution of the equation above is $c \cdot e^{-t^2}$, where $t(t)$ is the integral of $2t$. So $c \cdot e^{-t^2}$ is the solution, indeed $(c \cdot e^{-t^2})' + 2t \cdot c \cdot e^{-t^2} = 0$.

$$c \cdot -2t \cdot e^{-t^2} + 2t \cdot c \cdot e^{-t^2} = 0$$

$$0 = 0.$$

Integrating $c \cdot e^{-t^2}$ we get $\frac{c\sqrt{\pi}}{2} \operatorname{erf}(t) + c_2$ which we may simply write as $c_1 \operatorname{erf}(t) + c_2$ where erf is the error function.

26 april 2020

Gracim Ioan Flavin

912

Page 3

$$3. \quad \dot{x} = -y^2 - 4x$$

$$\dot{y} = xy + x^2$$

a) The equilibrium points are the solutions of $\begin{cases} -y^2 - 4x = 0 \\ xy + x^2 = 0 \end{cases}$

$$-y(y+x) = 0$$

$$x(y+x) = 0$$

$\Rightarrow$ either $x = y = 0$, or $x = -y$. So there

are an infinity of equilibria.

Now we may find a first integral by separating the variables and integrating

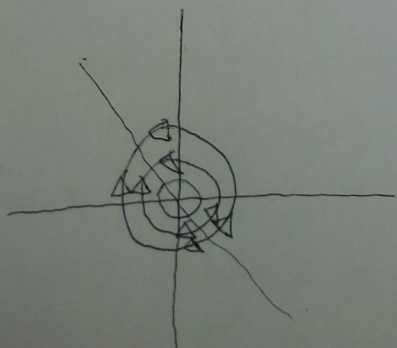
$$\frac{x'}{y'} = \frac{-y(y+x)}{x(y+x)}$$

$$x'x = -y'y$$

$$\int x'x = \int -y'y$$

$$\frac{x^2}{2} = -\frac{y^2}{2} + C$$

$x^2 + y^2 = C \Rightarrow$ The level curves of the first integral are circles which do not pass through $y = -x$



The straight line is the line which contains all equilibria

26 June 2020

Gracium van-Elwin Page 4

912

3. b) Looking at the phase portrait we notice arrows on the arcs point to the upper ~~right~~ left equilibria, so

$\lim_{t \rightarrow \infty} \varphi(t, 3, 0) = (-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}})$ because that point is the intersection of the circle and the line $x+y=0$ at the upper left point.

Similarly $\lim_{t \rightarrow \infty} \varphi(t, 1, 0) = (-\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}})$

c) These two functions are not periodic because they have a limit towards ∞ .

26 april 2020

Bräcium ban-Flareu Page 5

912

$$4. \quad \dot{x} = -\frac{1}{20}(x-10)$$

$$\dot{x} + \frac{1}{20}x - \frac{1}{2} = 0$$

First we solve the homogeneous part $\dot{x} + \frac{1}{20}x = 0$

characteristic polynomial $\lambda + \frac{1}{20} = 0 \Rightarrow \lambda = -\frac{1}{20} \Rightarrow c \cdot e^{-\frac{1}{20}t}$ is a solution to the homogeneous system.

A particular solution is a constant x so $\frac{1}{20}x = \frac{1}{2} \Rightarrow x = 10$

$c \cdot e^{-\frac{1}{20}t} + 10$ - the general solution

$$\text{IVP: } c \cdot e^{-\frac{1}{20} \cdot 0} + 10 = 4$$

$$c + 10 = 4$$

$$c = 4 - 10$$

The flow is $(4-10)e^{-\frac{1}{20}t} + 10$

To cool from 80° to 40° we solve

$$(80-10)e^{-\frac{1}{20}t} + 10 = 40$$

$$e^{-\frac{1}{20}t} = \frac{30}{70}$$

$$-\frac{1}{20}t = \ln \frac{3}{7}$$

$$t = 20 \ln \frac{7}{3}$$

$$5, a) \quad x_{k+1} = -x_k + 3y_k$$

$$y_{k+1} = -3x_k - y_k$$

$$y_k = \frac{x_{k+1} + x_k}{3}$$

$$\frac{x_{k+2} + x_{k+1}}{3} = -3x_k - \frac{x_{k+1} + x_k}{3} \quad | \cdot 3$$

$$x_{k+2} + x_{k+1} + x_{k+1} + 10x_k = 0$$

$$x_{k+2} + 2x_{k+1} + 10x_k = 0$$

$$x_0 = 0$$

$$y_0 = 2 \Rightarrow x_1 = 0 + 6 = 6$$

$$r^2 + 2r + 10 = 0$$

$$r_{1,2} = \frac{-2 \pm \sqrt{-36}}{2} = -1 \pm i\sqrt{3}$$

$$r_{1,2} = 2 \cdot (\cos \pm 120^\circ + i \sin \pm 120^\circ)$$

$$= 2 \left(\cos \pm \frac{2\pi}{3} \pm i \sin \pm \frac{2\pi}{3} \right)$$

$$x_k = C_1 \cdot 2 \cos \frac{2k\pi}{3} + C_2 \cdot 2 \sin \frac{2k\pi}{3}$$

$$x_0 = 0 \Rightarrow C_1 = 0$$

$$x_1 = 6 \Rightarrow C_2 \cdot 2 \cdot \frac{\sqrt{3}}{2} = 6 \Rightarrow C_2 = 2\sqrt{3}$$

26 iunie 2020

Grădinar (sen-Flavia)

912

Page 6

26. April 2020

Präcisionsbau-Flasche

912

Page 7

$$5. a) \text{ So } x_k = 2\sqrt{3} \cdot 2^k \cdot \sin \frac{2k\pi}{3}$$

$$x_k = \frac{y_{k+1} + y_k}{-3}$$

$$\frac{y_{k+2} + y_{k+1}}{-3} = \frac{y_{k+1} + y_k}{3} + 3y_k \quad | \cdot (-3)$$

$$y_{k+2} + y_{k+1} + y_{k+1} + y_k + 9y_k = 0$$

$$y_{k+2} + 2y_{k+1} + 10y_k = 0$$

$$y_k = c_1 \cdot 2^k \cos \frac{2k\pi}{3} + c_2 \cdot 2^k \sin \frac{2k\pi}{3}$$

$$y_0 = 2 \Rightarrow c_1 = 2$$

$$y_1 = -3 \cdot 0 - 2 = -2 \Rightarrow 2 \cdot 2 \cos \frac{2\pi}{3} + c_2 \cdot 2 \cdot \sin \frac{2\pi}{3} = -2$$

$$-2 + c_2 \cdot 2 \sin \frac{2\pi}{3} = -2 \Rightarrow \boxed{c_2 = 0}$$

$$\begin{cases} y_k = 2 \cdot 2^k \cdot \cos \frac{2k\pi}{3} \\ x_k = 2\sqrt{3} \cdot 2^k \sin \frac{2k\pi}{3} \end{cases}$$

solution of the IVP

Präcinn kan - Flaccin

912
26.11.2020 Page 8

$$S. 6) A = \begin{pmatrix} -1 & 3 \\ -3 & -1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} -\frac{1}{10} & \frac{3}{10} \\ \frac{3}{10} & -\frac{1}{10} \end{pmatrix}$$

$$A^{-2} = \begin{pmatrix} -\frac{2}{25} & \frac{3}{50} \\ \frac{3}{50} & -\frac{2}{25} \end{pmatrix}$$

AS Notice the terms keep getting smaller, approaching 0 because they are smaller than 1 by a lot and multiplying 2 terms smaller than 1 we get even smaller numbers.

$$\text{So } \lim_{n \rightarrow \infty} A^{-n} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Another solution is noticing $A^{-n} \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ is the solution for the IVP calculated at a) for $k = -n$, so $\lim_{n \rightarrow \infty} \begin{pmatrix} \lim_{n \rightarrow \infty} 2 \cdot 2^n \cos \frac{2n\pi}{3} \\ \lim_{n \rightarrow \infty} 2\sqrt{3} \cdot 2^n \sin \frac{2n\pi}{3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

6. ~~$f(2) = -5$~~

Suppose we start at $x = 2$

$f(2) = -5$. Now we are at $x = -5$

$f(-5) = 2$. We notice it is a cycle because 2 goes right back to -5 and -5 to 2 and so on.

We have established $\{ -5, 2 \}$ is a cycle. We also know a cycle is an attractor if $|f \circ f'(x)| < 1$.

$$x = 2$$

$$|(f \circ f)'(2)| = |f'(f(2)) f'(2)| = |f'(-5) f'(2)|$$

$$x = -5$$

$$|(f \circ f)'(-5)| = |f'(f(-5)) f'(-5)| = |f'(2) \cdot f'(-5)|$$

We have shown now that $|f \circ f'(x)| < 1$ and

$|f'(-5) f'(2)| < 1$ are equivalent.

for $x = 2$ or $x = -5$

Gracie van Elteren

912

26 June 2020

Page 9